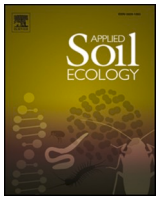
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Sustainable soil management: Insights from a 47-year maize-wheat cropping system study

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ABSTRACT

Continuous monitoring of soil quality is vital to understand the effects and to identify best management practices for sustainable agroecosystems. This study focuses on evaluating the soil quality and crop yield sustainability in a long-term maize-wheat cropping system experiment spanning 47 years. Physical, chemical, and biological soil quality indicators were assessed at two soil depths (0–0.15 m and 0.15–0.30 m) under eleven fertilizer and amendment regimes i.e., 50 % NPK, 100 % NPK, 150 % NPK, 100 % NPK + Zn, 100 % NPK+ hand weeding, 100 % NP, 100 % N, 100 % NPK + farmyard manure (FYM), 100 % NPK(—S), 100 % NPK + lime and control, laid out in a completely randomized block design. Principal component analysis (PCA) was employed to select a minimum data set (MDS) of indicators for soil quality assessment, followed by weighted additive approach to calculate the soil quality index (SQI). Results showed that balanced dose of NPK fertilizers, in combination with FYM or lime, enhanced average wheat grain yield by 39.9 and 29.4 % over sole 100 % NPK fertilizers, respectively. Treatment 100 % NPK + FYM recorded a significant improvement in aggregate stability, available nutrients, microbial biomass over control and 100 % NPK. The highest value of SQI (0–0.15 m) was recorded in 100 % NPK + FYM (0.99), followed by 100 % NPK + lime (0.86) and lowest in 100 % N (0.41). Omission of NPK, PK, K and S from fertilizer regime caused a decline of respective 33, 39, 22 and 6 % in the SQI, whereas, all the treatments recorded a decline in SQI in sub-surface layer. The average contribution of MDS towards the SQI of 0–0.15 and 0.15–0.30 m depth was labile carbon (LC) (30 %) > microbial biomass N (MBN) (28 %) > mean weight diameter (MWD) (17 %) > pH (16 %) > available K (6 %) and LC (33 %) > N mineralization (24 %) > MWD (22 %) > pH (21 %), respectively. The SQI positively correlated with average grain yield and SYI, highlighting the importance of soil quality for sustainable crop production. This research underscores the significance of continuous soil quality monitoring and use of balanced fertilization, organic manure and lime in maintaining soil health and crop productivity in agroecosystems.

1. Introduction

Soil quality, vis-à-vis soil health index (Chandel et al., 2018) is the ability of soil to sustain plant and animal productivity, maintain or enhance water and air quality, and support human health and habitation within natural or managed ecosystem (Mukherjee and Lal, 2014; Zhou et al., 2022). Both inherent and dynamic attributes characterize the soil quality, with the latter being notably influenced by land use management practices (Ji et al., 2020). Therefore, continuous monitoring of soil quality under agroecosystems plays a vital role in safeguarding the ecosystems, improving food quality, enhancing agricultural productivity, and mitigating land degradation (Chandel et al., 2018). Direct measurement of soil quality is challenging, leading to reliance on inferences drawn from changes in its biological, chemical, and physical attributes, often referred to as soil quality indicators (Bucher, 2002; Erkossa et al., 2007). Typically, certain sensitive indicators, also termed as minimum data set (MDS), are selected for soil quality assessment to establish correlations with key soil functions, such as crop yield (Thakur et al., 2022a). A contemporary approach to soil quality assessment is the concept of 'soil quality indexing' which entails normalizing data from measurements and converting it into a numeric value that surpasses a static descriptor, referred to as the soil quality index (SQI) (Andrews

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et al., 2004). Soil quality indexing enables the evaluation of a soil's ability to perform specific functions pertaining to physical, chemical and biological activities, serving as a foundation for managing and sustaining soil health.

Maize-wheat is one of the predominant cropping systems in the north-western Himalayan region (Rajneesh et al., 2017). Both crops are highly exhaustive and thus require high amounts of fertilizer nutrients (Gourav et al., 2018). Compared to traditional varieties, high-yielding varieties produce higher biomass, thereby depleting a significant amount of macronutrients as well as micronutrients from the soil (Majhi et al., 2019). Over the years, even with sufficient chemical fertilizer application, cereal-based cropping systems start to show stagnation or decline in yields (Prasad, 2005; Ram et al., 2016). Numerous studies have reported that neglecting organic inputs and prolonged chemical fertilizer use result in a decline in organic matter content (Singh, 2007; Williams et al., 2013; Ning et al., 2017), secondary and micronutrient deficiencies in the soil which ultimately leads to a stagnation/decline in soil fertility and crop productivity (Saha et al., 2018; Thakur et al., 2022b). There is an urgent need to shift the focus towards the health of cultivated soils so that ever-increasing demand for food can be met (Yang et al., 2015). The long-term productivity of agricultural systems with minimum impact on the environment is only possible with healthy soil. Continuous cropping and management practices, including the use of chemical fertilizers, manures and amendments, greatly affect the soil quality and the system's sustainability (Thakur et al., 2023).

The sustainability of a cropping system can be effectively determined using the 'sustainable yield index' (SYI) approach (Singh et al., 1990a; Choudhary et al., 2018). The SYI is a function of actual yields over a long period, and a high SYI indicates the higher capability of management practices for producing better yields over the years (Ram et al., 2016). The biological significance of soil quality indicators is validated through a significant correlation between SQI and crop yield (Li et al., 2013a). The yield trends and changes in soil health can be conveniently monitored in long-term fertilizer experiments (Saha et al., 2022). Long-term fertilizer experiments are invaluable tools for studying nutrient dynamics and identifying the factors associated with changes due to fertilizer use (Shahid et al., 2016; Choudhary et al., 2018). The complex interactions involving soil, plants, climate and management practices, along with their effects on the yield sustainability of crops, can be investigated through these experiments. Several researchers from various locations have assessed the soil quality index under long-term fertilization using various indexing methods (Mbuthia et al., 2015; Sharma et al., 2016; Zhou et al., 2022). Since soil quality assessment is purpose and site-specific, the indicators selected under MDS are different for different regions (Shukla et al., 2006). Dutta et al. (2015) selected pH, labile carbon (LC), available P (avl. P), available K (avl. K), diethylene triamine penta acetic acid (DTPA) extractable Zn, saturated hydraulic conductivity (SHC) and nitrogen mineralization (NM) for the MDS in silty loam soil of Palampur (India). Ram et al. (2016) chose mean weight diameter (MWD), bulk density (BD), microbial biomass carbon (MBC), dehydrogenase activity (DHA), organic carbon (OC), available N (avl. N), avl. P, avl. K and DTPA-extractable Zn for quality indexing of a silty clay loam soil under rice-wheat cultivation at Pantnagar (India), whereas Zhou et al. (2022) included nitrate nitrogen (NN), soil organic matter (SOM), pH, phosphatase activity (PA) and BD of a Mollisol of Hailun (China) in the mean data set, under maize-soybean rotation. Therefore, identifying MDS parameters can be useful for periodic monitoring of soil quality which in turn can help in optimizing nutrient application strategy. Limited information is available on the influence of cropping and fertilization application for more than four decades on soil quality in north-western Himalayan region.

A long-term fertilizer experiment was set up to investigate the effect different fertilization strategies on soil quality and crop yield. The objectives of present study were (1) to evaluate physical, chemical and biological soil properties, (2) to determine the sustainable yield index of wheat, (3) to work out the soil quality indices under continuous application of fertilizers and amendments over a period of 47 years.

Table 1
Some initial physico-chemical characteristics of the surface soil in 1972.

Soil property	Characteristic
Bulk density (Mg m^{-3})	1.31
Permeability	Moderately rapid
Mechanical separates (%)	
Sand	29.5
Silt	46.8
Clay	23.7
Textural class	Silt loam
Soil reaction	Acidic (pH 5.8)
Organic carbon (g kg^{-1})	7.9
Available macronutrients (kg ha^{-1})	
Nitrogen (N)	736
Phosphorus (P)	12.0
Potassium (K)	194
Exchangeable cations [c mol (p^{+}) kg^{-1}]	
Calcium (Ca)	5.30
Magnesium (Mg)	1.30
DTPA extractable micronutrients (mg kg^{-1})	
Iron (Fe)	26.0
Manganese (Mn)	24.3
Zinc (Zn)	1.9
Copper (Cu)	0.4

Table 2
Nutrients application regime (N:P:K, kg ha^{-1}) in different treatments.

Treatment	Maize		Wheat	
	Mineral fertilizers	FYM	Mineral fertilizers	FYM
50 % NPK	60:13:12.5	50.5:13:20	60:13:12.5	0:0:0
100 % NPK	120:13:25	0:0:0	120:13:25	0:0:0
150 % NPK	180:19.5:25	0:0:0	180:19.5:25	0:0:0
100 % NPK+ HW	120:26:25	0:0:0	120:26:25	0:0:0
100 % NPK+ Zn	120:26:25	0:0:0	120:26:25	0:0:0
100 % NP	120:26:0	0:0:0	120:26:0	0:0:0
100 % N	120:0:0	0:0:0	120:0:0	0:0:0
100 % NPK+ FYM	120:26:25	50.5:13:20	120:26:25	0:0:0
100 % NPK(—S)	120:26:25	0:0:0	120:26:25	0:0:0
100 % NPK+ Lime	120:26:25	0:0:0	120:26:25	0:0:0
Control	0:0:0	0:0:0	0:0:0	0:0:0

2. Material and methods

2.1. Experimental site, layout and treatments

The field study was conducted in a 47-year-old long-term fertilizer experiment (LTFE) involving the maize-wheat cropping sequence. Initiated in 1972, the LTFE took place at the experimental farm of the Department of Soil Science, Chaudhary Sarwan Kumar Himachal Pradesh Agricultural University, Palampur, India (31^0 6′ N, 76^0 3′ E, 1290 m above sea level). The region experiences a humid sub-temperate climate characterized by mild summers, severe winters and an annual rainfall exceeding 2500 mm. Throughout the crop growth period, the air temperature ranged from 1 °C to 31 °C, and received a total of 496 mm of rainfall. The experimental site's soil is characterized as silty loam and exhibits an acidic reaction (Table 1). This soil is classified under the Typic Hapludalf sub-group according to the Soil Survey Staff (1975).

The experiment followed a completely randomized block design with three replications for eleven treatments (Table 2), each implemented in plots measuring 5 m × 3 m. The 100 % NPK dose represents the state-level fertilizer recommendation of 120 kg N, 60 kg P_2O_5 and 40 kg ha^{-1} K_2O for maize and 120 kg N, 60 kg P_2O_5 and 30 kg ha^{-1} K_2O for wheat. Beginning in June 2011, the P_2O_5 doses in 100 % NPK and 150 % NPK treatments were halved due to a marked build-up of available P.

Also, farmyard manure (FYM) application (to maize only) commenced in 50 % NPK treatment. Nitrogen was supplied via urea in three split doses i.e., half at the time of sowing and the remaining half in two equal splits at the 8-leaf and pre-tasselling stages of maize or maximum tillering and flowering stages of wheat. Full phosphorus and potassium doses were incorporated at sowing through single super phosphate (SSP) and muriate of potash (MOP). However, in the 100 % NPK(—S) treatment, phosphorus was applied through diammonium phosphate (DAP) to assess the impact of sulphur-free phosphorus fertilizer. In 100 % NPK + Zn treatment, 25 kg Zn ha^{-1} was added annually through $ZnSO_4$ to both crops until 2011. In 100 % NPK + lime treatment, 900 kg lime ha^{-1} ($CaCO_3$ passed through 100 mesh sieve) was applied each year before maize sowing to elevate soil pH to 6.5, but after 1978, it was applied only when the soil pH would fall below 6.3. In 50 % NPK and 100 % NPK + FYM treatments, well-decomposed FYM (60 % moisture, 1.01 % N, 0.26 % P and 0.40 % K) was added at the rate of 5 t ha^{-1} based on dry weight to maize crop only.

Each plot was tilled separately to a depth of 0.15 m using a power tiller. After field preparation, wheat (variety HPW-368) was sown on 17th Nov 2018 with a row spacing of 20 cm. One pre-sowing irrigation was followed by subsequent irrigations at the crown root initiation, tillering, late jointing, flowering and dough stages. After seed emergence, active ingredients of Isoporoturon were applied at a rate of 1.125 kg ha^{-1} to control weeds in all treatments, except for the 100 % NPK + HW. In this treatment, weeds were manually uprooted and incorporated into the respective plots. The crop was harvested on 10th May 2019.

2.2. Soil sampling and analysis

Immediately after harvesting, soil samples were taken from 0 to 0.15 and 0.15–0.30 m soil depths using a core from each plot. One part of the moist soil sample was sieved through an 8 mm sieve and stored in plastic bags at 4 °C for biological analysis. The MBC and microbial biomass nitrogen (MBN) were determined with the fumigation-extraction method as described by Vance et al. (1987) and Jenkinson and Ladd (1981), respectively. The NM was estimated through anaerobic incubation and KCl extraction, described earlier by Waring and Bremner (1964) and later modified by Keeney (1982). The remaining part of the soil sample was air-dried, ground with a wooden mortar pestle and passed through a 2 mm sieve for physical and chemical analysis. A core sampler was used to measure BD (Singh, 1980) and the mean weight diameter (MWD) of water-stable aggregates was determined by the wet sieving method (Yoder, 1936). The constant head method was followed to determine SHC (Klute, 1965). Soil pH and electrical conductivity (EC) of soil-water suspensions of 1:2.5 and 1:2 (*w/v*) were measured using a compound electrode (Jackson, 1973), respectively. Soil organic carbon was determined by potassium dichromate oxidation and titration of excess dichromate against ferrous ammonium sulphate (Walkley and Black, 1934). Labile C content of the soil was determined as per Weil et al. (2003). Available N was determined using alkaline potassium permanganate and Kjeldahl assembly (Subbiah and Asija, 1956). Bray no. 1 reagent (Bray and Kurtz, 1945) was used to extract available P, followed by spectrophotometric estimation through the ascorbic acid method. Available K was extracted using a neutral normal ammonium acetate solution through the displacement of exchangeable cations and measured with a flame photometer (Jackson, 1973). Micronutrient cations (Fe, Mn, Zn and Cu) were extracted with DTPA solution and estimated by an atomic absorption spectrophotometer (Lindsay and Norvell, 1978).

2.3. Soil quality index

Soil quality index was determined by following four main steps, based on the method described by Andrews et al. (2002a, b). Firstly, an objective was defined, then MDS was selected which was relevant to the soil functions, followed by linear scoring of the MDS indicators and then the indicators were integrated into an index. In the present study, wheat yield sustainability was chosen as the objective because it is vital to maintain the high yields obtained from good management practices. Wheat yield sustainability was determined in terms of SYI, which signified the minimum guaranteed yield in response to a nutrient management treatment as a percentage of the maximum observed yield with high probability. It was calculated based on wheat yield obtained in previous years from 1972 to 73 to 2018–19 for each treatment, using the following formula given by Singh et al. (1990b):

$$SYI = \frac{\bar{y} - \sigma}{Ymax}$$

where $\bar{y}$ is the estimated average yield of a treatment, σ is the estimated standard deviation and Y_{max} is the maximum observed yield in the experiment over the years. If the standard deviation (σ) is low, SYI will be high, indicating a sustainable management practice and vice-versa (Shahid et al., 2013).

All the analysed soil properties at 0–0.15 m and 0.15–0.30 m soil depths, except EC, were subjected to principal component analysis (PCA). Since the soil under study was acidic and EC values under all the treatments were very less (below 66.30 μS cm^{-1}), EC was excluded from PCA. The principal components (PC) with eigenvalues >1 (Kaiser, 1960) which explained at least 5 % of the variation in the data were taken into consideration (Wander and Bollero, 1999). Within the chosen PC, variables having factor loading values within 10 % of the highest factor loading were selected and subjected to multivariate correlation analysis. Among the well-correlated variables, one with the highest factor loading value was selected. Expert opinion and correlation sum methods were also employed to select variables for the MDS. After selecting the MDS, each variable was transformed into an indicator score (S), ranging from 0 (least favourable) to 1 (most favourable), using the linear scoring method (Andrews et al., 2002a). The indicators were categorized depending on whether “more is better” or “less is better”. Since all the selected parameters in MDS were good in increasing order, they were scored as “more is better”, i.e., the recorded values were divided by the highest value so that the highest value got the score of 1 and others got <1. After the transformation, the variables were integrated into indices using a weighted additive approach (Mukherjee and Lal, 2014). The variation (%) in the dataset, explained by each PC, was divided by the total variation percentage explained by all PCS (eigenvalues >1) to obtain the weightage value under a particular PC. Thereafter, the SQI was calculated by adding up the weighted MDS scores using the following equation:

$$SQI = \sum_{i=1}^{n} W_i S_i$$

where S_i is the score of the variable and W_i is the weight of the variable derived from PCA. We assumed that a higher score meant higher soil quality and higher crop yield sustainability. The average of weightage value of 11 treatments of an MDS indicator was calculated to work out its per cent contribution towards SQI. The SQI was also validated against average grain yield and SYI by performing correlation analysis (Mukherjee and Lal, 2014).

2.4. Statistical analysis

The experimental data were subjected to analysis of variance (ANOVA) and the treatment means were compared using Duncan's multiple range tests ($p = 0.05$) using OPSTAT (Sheoran et al., 1998). The PCA was performed in SPSS version 23 and PCA biplots were obtained from PAST version 4.03. Bar diagrams and linear regression graphs were prepared in Microsoft Excel 2019.

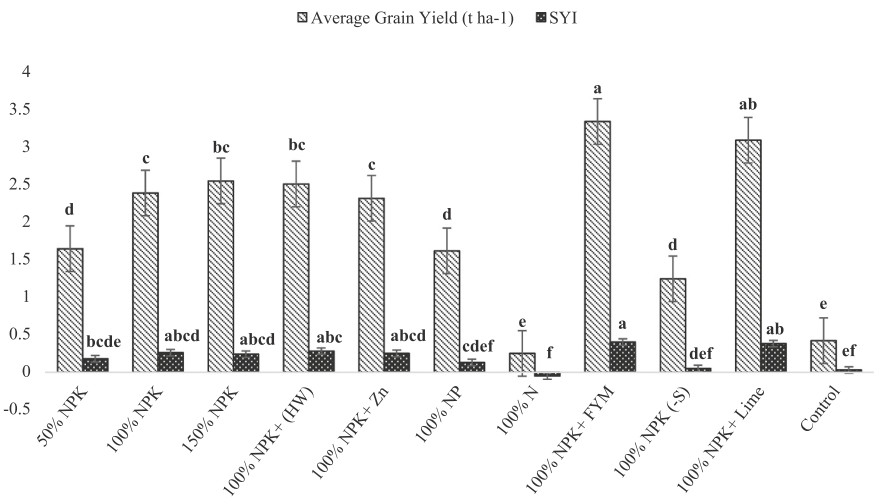


Fig. 1. Effect of long-term fertilizers and amendments application on average grain yield and sustainable yield index (SYI) from 1972 to 2019. Bars represent standard error. Values with different lowercase letters within a column are significantly different according to DMRT (p = 0.05).

3. Results

3.1. Wheat yield and SYI

Application of fertilizers, either alone or in combination with FYM or lime enhanced the average (47 years average) grain yield of wheat significantly over control, barring 100 % N (Fig. 1). The maximum average grain yield (3.33 t ha^{-1}) was recorded in 100 % NPK + FYM treatment and the lowest (0.25 t ha^{-1}) in 100 % N. It was observed that the inherent soil nutrients (control) were capable of sustaining 0.42 t ha^{-1} of wheat grain yield. A great decline (68 %) in the average grain yield was noted in the 100 % N treatment as compared to the control. With the balanced application of NPK fertilizers, the average grain yield was sustained at 2.38 t ha^{-1}. However, the conjoint use of balanced NPK fertilizers and FYM generated a 39.9 % higher yield than 100 % NPK alone. Furthermore, lime application in 100 % NPK + lime enhanced the grain yield by 29.4 % as compared to 100 % NPK. The SYI for wheat varied significantly within different treatments. The SYI was found to be highest (0.40) in 100 % NPK + FYM, followed by 100 % NPK + lime (0.38) and lowest (−0.05) in 100 % N treatment. Control with no added fertilizers recorded higher SYI (0.03) than 100 % N. Balanced application of NPK fertilizers increased the SYI by 81 and 50 % as compared to imbalanced fertilizer application in 100 % NPK (—S) and 100 % NP treatments. An increase in the NPK doses from 100 to 150 % depleted the SYI by 8 %.

3.2. Soil quality indicators

The mean values of all physical, chemical and biological properties at 0–0.15 m and 0.15–0.30 m soil depths are given in Tables 3 and 4, respectively. The BD at 0–0.15 m soil depths varied from 1.18 Mg m^{-3} in 100 % NPK + FYM to 1.40 Mg m^{-3} in 100 % N, respectively. The respective MWD values were 4.84 and 0.89 mm. The SHC was found to be highest under 100 % NPK + FYM (5.44 cm hr^{-1}), followed by 100 % NPK + lime (4.16 cm hr^{-1}) and lowest under control (2.50 cm hr^{-1}). Farmyard manure addition with 100 % NPK resulted in a 7 % decline in BD while a 52 and 22 % increase in MWD and SHC, compared to sole 100 % NPK, respectively. A marked decline in the soil pH from an initial value of 5.8 was recorded in all the treatments, except 100 % NPK + lime (pH 6.19). The extent of the decline was maximum in 100 % N (pH 4.39) and lowest in 100 % NPK + FYM (pH 5.72). A significant effect of the fertilizers and amendments application on soil EC was noted. The EC was significantly higher under 150 % NPK (66.3 μS cm^{-1}) as compared to all other treatments except 100 % N (63.8 μS cm^{-1}). Nevertheless, EC values in all the treatments were within the safe limit.

Continuous fertilizers and amendments application significantly altered the soil OC content. Maximum OC (13.5 g kg^{-1}) was recorded in 100 % NPK + FYM, followed by 100 % NPK + HW (11.6 g kg^{-1}) and the lowest in control (8.2 g kg^{-1}). The SOC increased by 4 to 71 % under continuous cropping from the initial value (7.9 g kg^{-1}), the increase being highest in 100 % NPK + FYM. All the fertilizer treatments, except 100 % N, significantly enhanced the LC content over the unfertilized control. The FYM and lime application in 100 % NPK + FYM and 100 % NPK + lime treatment augmented the LC to the tune of 29.6 and 12.0 % over 100 % NPK, respectively. The N availability showed a considerable depletion under all treatments as compared to the initial value (736 kg ha^{-1}) but the extent of decline was maximum in control (63 %). The N availability in 100 % NPK + FYM (389 kg ha^{-1}) was highest and statistically comparable to 100 % NPK + HW (373 kg ha^{-1}). On the contrary, available P content increased from 12.0 kg ha^{-1} (initial) in all the treatments. Treatment 150 % NPK recorded the highest P build-up with 156 kg ha^{-1} available P content. In the absence of P fertilizer application, its availability declined by 79.2 and 80.6 % in 100 % N and control as compared to the balanced 100 % NPK application. A substantial reduction in K availability was recorded in all the treatments over the initial content (194 kg ha^{-1}), except 100 % NPK + FYM (201 kg ha^{-1}). The available K depletion was maximum in control (39.2), closely followed by 100 % N (35.6) and 100 % NP (35.0 %).

The DTPA-extractable micronutrients were found to be sufficient in all the treatments, however, a marked decline from the initial content was noted. Significantly higher contents of DTPA-extractable Fe, Mn and Cu were recorded in 100 % NPK + FYM than the rest of the treatments, but Zn was maximum in 100 % NPK + Zn. In contrast, the control recorded the lowest DTPA-extractable micronutrients. The application of balanced fertilizers with FYM and lime exhibited a positive influence on soil biological properties. Microbial biomass C and N, and NM were significantly enhanced in all fertilizer treatments over control, except in 100 % N. Treatment 100 % NPK + FYM recorded the highest MBC (646 mg kg^{-1}), MBN (25.0 mg kg^{-1}) and NM (33.2 mg kg^{-1}), and was statistically superior to all other treatments. Hand weeding and weed incorporation in 100 % NPK + HW treatment significantly improved the MBC and MBN over 100 % NPK to the tune of 20.2 and 8.2 %, respectively. Moving down the soil profile (0.15–0.30 m), BD and pH tended to increase whereas the rest of the soil properties decreased under all the treatments. However, the treatment-wise trend was analogous to the surface layer (0–0.15 m).

3.3. Selection of MDS

Taking 16 physical, chemical and microbiological soil attributes, PC

Table 3
Effect of long-term fertilizers and amendments application on selected soil quality indicators (0–0.15 m). Means followed by different lowercase letters within a column are significantly different according to DMRT (p = 0.05).

Treatment	BD (Mg m^{-3})	MWD (mm)	SHC (cm hr^{-1})	pH	EC (μS cm^{-1})	OC (g kg^{-1})	LC (g kg^{-1})	Avl. N (kg ha^{-1})	Avl. P (kg ha^{-1})	Avl. K (kg ha^{-1})	DTPA extractable micronutrients (mg kg^{-1})				MBC (mg kg^{-1})	MBN (mg kg^{-1})	NM (mg kg^{-1})
											Fe	Mn	Zn	Cu			
50 % NPK	1.27^{ef}	1.69^{e}	3.24^{f}	5.29^{de}	51.8^{f}	10.5^{cd}	1.29^{e}	347^{de}	57^{h}	151^{d}	28.4^{cd}	21.4^{de}	1.33^{cd}	1.75^{b}	380^{f}	14.9^{g}	16.5^{e}
100 % NPK	1.28^{de}	1.97^{d}	3.45^{de}	5.21^{de}	62.5^{ab}	10.2^{de}	1.42^{d}	365^{bcd}	72^{g}	162^{c}	32.7^{b}	22.5^{cde}	1.28^{cd}	1.73^{bc}	461^{d}	18.4^{d}	19.6^{c}
150 % NPK	1.29^{de}	1.58^{e}	3.47^{cde}	4.89^{f}	66.3^{a}	9.9^{de}	1.53^{bc}	369^{bc}	156^{a}	181^{b}	33.4^{ab}	25.3^{b}	1.34^{c}	1.72^{bc}	431^{e}	20.2^{c}	20.4^{c}
100 % NPK+ HW	1.25^{fg}	2.42^{c}	3.61^{c}	5.33^{cd}	60.8^{bcd}	11.6^{b}	1.45^{cd}	373^{ab}	88^{e}	163^{c}	32.3^{b}	24.1^{bcd}	1.38^{c}	1.81^{b}	554^{c}	19.9^{c}	20.2^{c}
100 % NPK+ Zn	1.27^{ef}	1.65^{e}	3.33^{ef}	5.20^{de}	57.1^{cde}	9.6^{e}	1.42^{d}	361^{bcd}	81^{f}	172^{bc}	26.9^{def}	21.6^{de}	4.19^{a}	1.76^{b}	430^{e}	17.0^{e}	19.8^{c}
100 % NP	1.32^{c}	1.54^{e}	2.80^{g}	5.12^{e}	62.0^{abc}	9.7^{e}	0.95^{f}	350^{cde}	104^{d}	126^{e}	27.6^{cde}	22.2^{cde}	1.31^{cd}	1.62^{d}	267^{g}	14.1^{g}	18.0^{d}
100 % N	1.40^{a}	0.89^{f}	2.66^{gh}	4.39^{g}	63.8^{ab}	8.5^{f}	0.76^{g}	333^{e}	15^{i}	125^{e}	31.0^{bc}	22.0^{cd}	1.38^{c}	1.50^{e}	176^{i}	11.5^{h}	9.5^{g}
100 % NPK+ FYM	1.18^{h}	4.84^{a}	5.44^{a}	5.72^{b}	54.7^{ef}	13.5^{a}	1.84^{a}	389^{a}	146^{b}	201^{a}	36.8^{a}	39.7^{a}	2.47^{b}	2.26^{a}	646^{a}	25.0^{a}	33.2^{a}
100 % NPK(—S)	1.30^{cd}	1.61^{e}	3.58^{cd}	5.16^{de}	56.1^{def}	9.8^{e}	1.43^{cd}	351^{cde}	130^{c}	173^{bc}	23.6^{f}	21.2^{e}	1.47^{c}	1.65^{cd}	436^{e}	16.0^{f}	19.0^{cd}
100 % NPK+ Lime	1.24^{g}	3.71^{b}	4.16^{b}	6.19^{a}	59.2^{bcde}	11.2^{bc}	1.59^{b}	358^{bcd}	79^{f}	167^{c}	24.2^{ef}	24.9^{bc}	1.34^{c}	1.78^{b}	618^{b}	22.0^{b}	22.9^{b}
Control	1.37^{b}	1.09^{f}	2.50^{h}	5.48^{c}	40.4^{g}	8.2^{f}	0.88^{f}	271^{f}	14^{i}	118^{e}	18.5^{g}	17.0^{f}	1.12^{d}	1.39^{f}	235^{h}	10.2^{i}	13.4^{f}

Table 4
Effect of long-term fertilizers and amendments application on selected soil quality indicators (0.15–0.30 m). Means followed by different lowercase letters within a column are significantly different according to DMRT (p = 0.05).

Treatment	BD (Mg m^{-3})	MWD (mm)	SHC (cm hr^{-1})	pH	EC (μS cm^{-1})	OC (g kg^{-1})	LC (g kg^{-1})	Avl. N (kg ha^{-1})	Avl. P (kg ha^{-1})	Avl. K (kg ha^{-1})	DTPA extractable micronutrients (mg kg^{-1})				MBC (mg kg^{-1})	MBN (mg kg^{-1})	NM (mg kg^{-1})
											Fe	Mn	Zn	Cu			
50 % NPK	1.30^{de}	1.55^{cd}	3.18^{c}	5.38^{d}	49.6^{e}	7.74^{c}	0.68^{d}	252^{cde}	41^{f}	141^{e}	20.5^{de}	17.9^{bc}	1.18^{c}	1.18^{d}	288^{e}	11.5^{d}	10.9^{f}
100 % NPK	1.31^{cd}	1.73^{c}	3.36^{c}	5.29^{de}	56.9^{abc}	7.55^{c}	0.78^{cd}	262^{bc}	63^{e}	150^{cd}	24.6^{c}	17.5^{bcd}	1.22^{c}	1.23^{c}	414^{c}	15.3^{b}	16.7^{b}
150 % NPK	1.32^{cd}	1.40^{cd}	3.24^{c}	5.05^{f}	61.0^{a}	7.49^{c}	0.91^{b}	265^{b}	130^{a}	160^{b}	27.9^{ab}	18.7^{b}	1.29^{c}	1.07^{ef}	396^{cd}	16.0^{b}	16.8^{b}
100 % NPK+ HW	1.26^{ef}	2.25^{b}	3.43^{c}	5.36^{d}	54.8^{bcd}	8.24^{b}	0.87^{bc}	269^{b}	75^{d}	147^{de}	24.0^{c}	18.3^{bc}	1.25^{c}	1.27^{b}	463^{b}	15.7^{b}	16.8^{b}
100 % NPK+ Zn	1.29^{de}	1.48^{cd}	3.21^{c}	5.24^{e}	52.2^{cde}	7.26^{c}	0.85^{bc}	258^{bcd}	72^{d}	143^{de}	23.5^{c}	17.6^{bc}	2.33^{a}	1.20^{cd}	384^{cd}	13.3^{c}	11.8^{e}
100 % NP	1.35^{c}	1.38^{cd}	2.56^{d}	5.21^{e}	56.2^{bc}	6.54^{d}	0.68^{d}	242^{e}	86^{c}	111^{f}	22.5^{cd}	16.9^{cd}	1.20^{c}	1.07^{ef}	230^{f}	11.8^{d}	13.3^{d}
100 % N	1.44^{a}	0.59^{e}	2.48^{d}	4.46^{g}	58.4^{ab}	6.08^{d}	0.46^{e}	251^{de}	8^{g}	107^{fg}	26.8^{b}	15.1^{e}	0.99^{d}	1.03^{f}	147^{g}	9.5^{e}	5.9^{h}
100 % NPK+ FYM	1.22^{f}	3.61^{a}	4.30^{a}	5.74^{b}	49.4^{e}	9.27^{a}	1.16^{a}	300^{a}	127^{a}	173^{a}	29.5^{a}	26.1^{a}	1.74^{b}	1.83^{a}	533^{a}	17.5^{a}	28.7^{a}
100 % NPK(—S)	1.32^{cd}	1.42^{cd}	3.26^{c}	5.21^{e}	51.2^{de}	6.39^{d}	0.84^{bc}	263^{b}	103^{b}	156^{bc}	18.9^{e}	16.2^{de}	1.19^{c}	1.09^{e}	351^{d}	12.2^{cd}	15.7^{c}
100 % NPK+ Lime	1.26^{ef}	3.17^{a}	3.89^{b}	6.25^{a}	55.2^{bcd}	7.60^{c}	0.93^{b}	268^{b}	61^{e}	160^{b}	23.1^{c}	13.1^{f}	1.27^{c}	1.23^{c}	502^{ab}	16.6^{ab}	16.9^{b}
Control	1.40^{b}	0.99^{de}	2.46^{d}	5.53^{c}	35.8^{f}	6.54^{d}	0.54^{e}	173^{f}	8^{g}	100^{g}	16.5^{f}	10.1^{g}	0.96^{d}	1.05^{ef}	184^{fg}	7.5^{f}	10.4^{g}

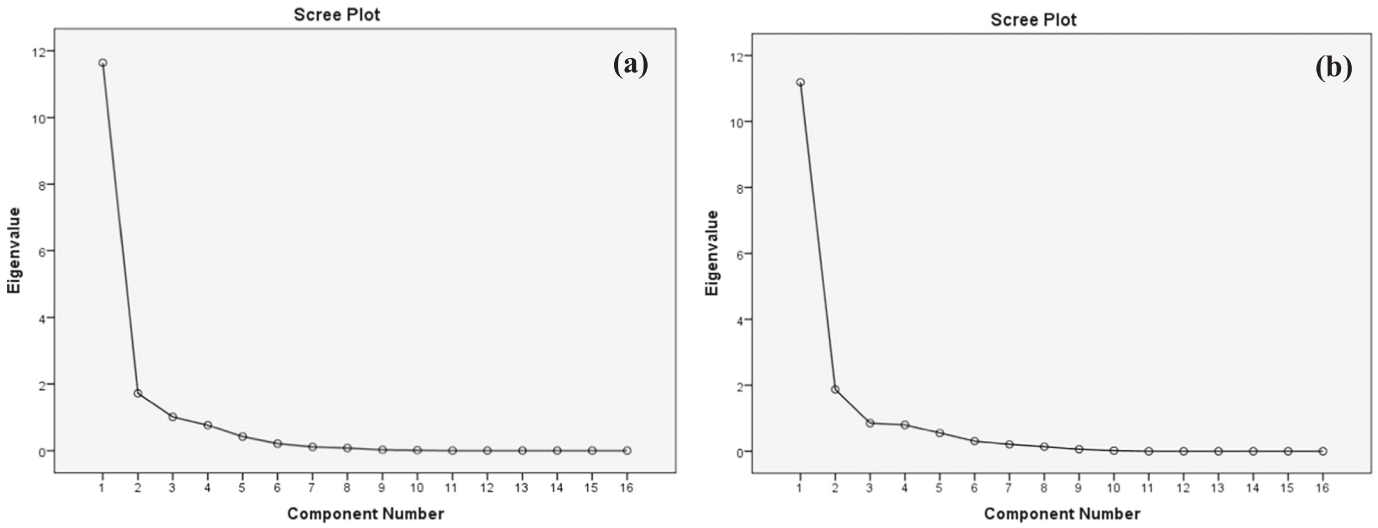


Fig. 2. Spree plot of principal component analysis for (a) 0–0.15 m and (b) 0.15–0.30 m soil depths.

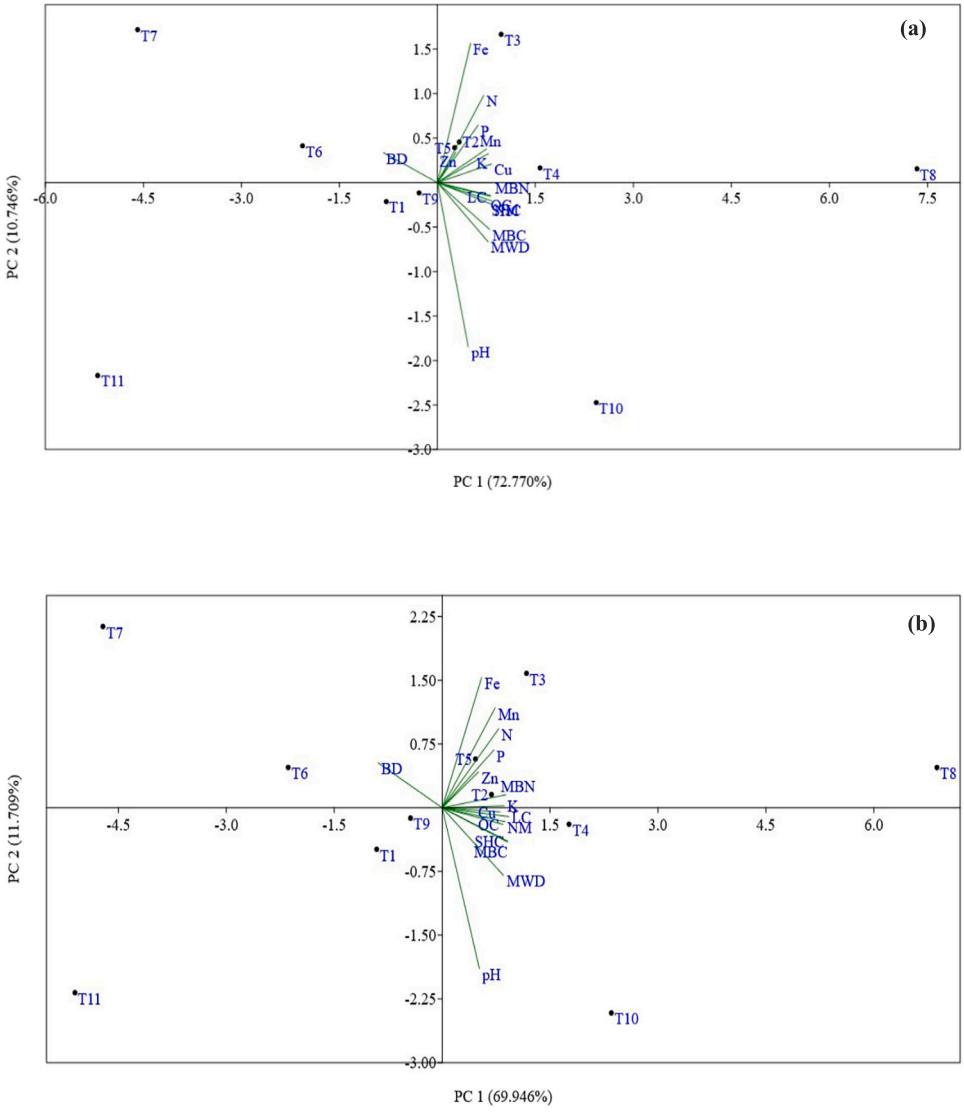


Fig. 3. Biplots of principal component analysis for (a) 0–0.15 m and (b) 0.15–0.30 m soil depths.

Table 5
Results from the Principal Components Analysis of soil quality indicators (0–0.15 m).

Principal components	PC_1	PC_2	PC_3
Eigen value	11.643	1.719	1.013
Percentage of variance	72.770	10.746	6.328
Cumulative variance Percentage	72.770	83.516	89.845
Weightage	0.810	0.120	0.070
Potential MDS variables	**Factor loading values**		
BD	**−0.955**	0.150	−0.076
SHC	**0.967**	−0.092	−0.058
MWD	**0.902**	−0.297	−0.165
pH	0.549	**−0.825**	−0.009
Avl. N	0.824	0.438	0.011
Avl. P	0.725	0.289	0.053
Avl. K	**0.906**	0.146	**0.224**
DTPA Fe	0.591	0.699	−0.280
DTPA Cu	**0.964**	0.093	−0.008
DTPA Zn	0.328	0.152	**0.866**
DTPA Mn	0.870	0.168	−0.193
OC	**0.949**	−0.064	−0.184
LC	**0.941**	−0.069	0.135
MBC	**0.930**	−0.235	0.020
MBN	**0.968**	0.000	−0.055
NM	**0.963**	−0.108	0.029

Boldface numbers were highly weighted variables under each principal component. The underlined variables were retained in the MDS.

Table 6
Results from the Principal Components Analysis of soil quality indicators (0.15–0.30 m).

Principal components	PC_1	PC_2
Eigen value	11.191	1.873
Percentage of variance	69.946	11.709
Cumulative variance Percentage	69.946	81.655
Weightage	0.857	0.143
Potential MDS variables	**Factor loading values**	
BD	**−0.933**	0.227
SHC	**0.960**	−0.170
MWD	**0.893**	−0.342
pH	0.543	**−0.814**
Avl. N	0.825	0.400
Avl. P	0.755	0.292
Avl. K	**0.905**	0.010
DTPA Fe	0.575	0.658
DTPA Cu	0.842	−0.021
DTPA Zn	0.528	0.181
DTPA Mn	0.773	0.505
OC	**0.888**	−0.084
LC	**0.971**	−0.045
MBC	**0.946**	−0.172
MBN	**0.928**	0.065
NM	**0.914**	−0.070

Boldface numbers were highly weighted variables under each principal component. The underlined variables were retained in the MDS.

analysis was performed for 0–0.15 and 0.15–0.30 m soil depths. For 0–0.15 m depth, three PCs were selected having eigenvalue >1 and these explained 85.5 % variation in the data (Table 4). Figs. 2(a) and 3 (a) show the scree plot and biplot of PCA for 0–0.15 m soil depth, respectively. The PC1, PC2 and PC3 explained respective 72.8, 10.7 and 6.3 % variance. From the principal component loading matrix, ten variables viz., BD, SHC, MWD, K, Cu, OC, LC, MBC, MBN and NM were found to be highly weighted variables in PC1. From these ten variables, three variables (MBN, MWD and LC) were selected (Table 5). The MBN was chosen because of its highest factor loading and correlation sum values (Supplementary Table 1). Among physical attributes, MWD was selected, whereas LC was selected from the chemical attributes. In PC2, pH was retained and in PC3, avl. K was selected. Therefore, the MDS for 0–0.15 m soil depth included MBN, MWD, LC, pH and avl. K. Linear scores were calculated through the "more is better" approach (Table 7). Then, the weight for PC1, PC2 and PC3 were calculated. In PC1, MBN, MWD and LC were well-correlated, so the weight (0.810) was divided equally among these three, as suggested by Singh et al. (2015), thus, resulting in a weight of 0.27 each. In PC2 and PC3, full weights of 0.12 and 0.07 were assigned to pH and avl. K. Therefore, the following equation was used to calculate SQI for 0–0.15 m soil depth:

$$SQI = (0.27 \times S_{MBN}) + (0.27 \times S_{MWD}) + (0.27 \times S_{LC}) + (0.12 \times S_{pH}) + (0.07 \times S_K)$$

where S is the score of the subscripted variable and the coefficient is the respective weight.

For 0.15–0.30 m soil depth, two PCs had >1 eigenvalues (Table 6). The spree plot and biplot of PCA for 0.15–0.30 soil depth are shown in Fig. 2(b) and 3(b). Almost 81.6 % of the data variability was explained by these PCs with a respective contribution of 69.9 and 11.7 % from PC1 and PC2. In PC1, nine variables were found to be highly weighted namely, BD, SHC, MWD, avl. K, OC, LC. MBC, MBN and NM. All these variables were strongly correlated (Supplementary Table 2). Out of these, LC was selected along with MWD (physical variable) and NM (biological variable). In PC2, the variable with the highest factor loading value i.e., pH was selected. Scores for all the retained variables were computed through linear scoring. The calculated weight of PC1 (0.857) was equally divided into three variables with a value of 0.28 each. In PC2, pH was allocated the full weight value of 0.14. Therefore, the SQI equation was as follows:

$$SQI = (0.28 \times S_{LC}) + (0.28 \times S_{MWD}) + (0.28 \times S_{NM}) + (0.14 \times S_{pH})$$

where S is the score of the subscripted variable and the coefficient is the respective weight.

3.4. Soil quality index

The SQI values varied from 0.41 to 0.99 for 0–0.15 m (Table 7) and 0.31 to 0.97 for 0.15–0.30 m soil depth (Table 8). It is evident that the application of FYM and lime in addition to the recommended NPK dose would improve the SQI of surface soil by 48 and 28 %, respectively. It was further observed that continuous cropping in the absence of NPK (control), PK (100 % N), K (100 % NP) and S {100 % NPK(—S)} would decline the soil quality by 33, 39, 22 and 6 %, respectively, as compared to the balanced fertilization in 100 % NPK. We found a marked improvement in the SQI of hand-weeded plots from chemical weed control (100 % NPK). The SQI declined with depth in all of the treatments. Average contribution of each MDS indicator towards the SQI for 0–0.15 m and 0.15–0.30 m soil depths has been depicted in Fig. 4(a) and 4(b), respectively. At 0–0.15 m soil depth, LC (30 %) made the highest contribution towards SQI, followed by MBN (28 %), MWD (17 %), pH (16 %) and avl. K (6 %). The order of contribution of variables towards SQI of 0.15–0.30 m soil depth was LC (33 %) > NM (24 %) > MWD (22 %) > pH (21 %). Soil quality indices of 0–0.15 m and 0.15–0.30 m soil depths were found to be significantly correlated with the average wheat grain yield ($R^2 = 0.853^{**}$ and 0.781^{**}), as well as the SYI ($R^2 = 0.835^{**}$ and 0.790^{**}), respectively (Fig. 5).

4. Discussion

4.1. Wheat yield and SYI

Sustainability refers to the enhancement and/or maintenance of productivity for a long time through integrated management practices. Singh et al. (1990a) proposed the SYI as a quantitative measure to assess the sustainability of an agricultural practice. It helps to establish the minimum guaranteed yield that can be obtained relative to the maximum observed yield. Owing to the high yields in the last 47 years, 100 % NPK + FYM recorded the highest SYI and AGY of wheat. The positive effect of FYM application on wheat yield may be due to its

Table 7
Score, weight and soil quality index (SQI) values of selected minimum data set (MDS) variables (0–0.15 m) for each treatment.

Treatments	MBN		MWD		LC		pH		Avl. K		SQI
	Score (S)	Weight (W)	Score (S)	Weight (W)	Score (S)	Weight (W)	Score (S)	Weight (W)	Score (S)	Weight (W)	
50 % NPK	0.60	0.27	0.35	0.27	0.70	0.27	0.85	0.12	0.75	0.07	0.60
100 % NPK	0.74	0.27	0.41	0.27	0.77	0.27	0.84	0.12	0.81	0.07	0.67
150 % NPK	0.81	0.27	0.33	0.27	0.83	0.27	0.79	0.12	0.90	0.07	0.69
100 % NPK+ (HW)	0.80	0.27	0.50	0.27	0.79	0.27	0.86	0.12	0.81	0.07	0.72
100 % NPK+ Zn	0.68	0.27	0.34	0.27	0.77	0.27	0.84	0.12	0.86	0.07	0.64
100 % NP	0.56	0.27	0.32	0.27	0.52	0.27	0.83	0.12	0.63	0.07	0.52
100 % N	0.46	0.27	0.18	0.27	0.41	0.27	0.71	0.12	0.62	0.07	0.41
100 % NPK+ FYM	1.00	0.27	1.00	0.27	1.00	0.27	0.92	0.12	1.00	0.07	0.99
100 % NPK (—S)	0.64	0.27	0.33	0.27	0.78	0.27	0.83	0.12	0.86	0.07	0.63
100 % NPK+ Lime	0.88	0.27	0.77	0.27	0.87	0.27	1.00	0.12	0.83	0.07	0.86
Control	0.41	0.27	0.22	0.27	0.48	0.27	0.89	0.12	0.59	0.07	0.45

Table 8
Score, weight and soil quality index (SQI) values of selected minimum data set (MDS) variables (0.15–0.30 m) for each treatment.

Treatments	LC		MWD		NM		pH		SQI
	Score (S)	Weight (W)	Score (S)	Weight (W)	Score (S)	Weight (W)	Score (S)	Weight (W)	
50 % NPK	0.59	0.28	0.43	0.28	0.38	0.28	0.86	0.14	0.51
100 % NPK	0.67	0.28	0.48	0.28	0.58	0.28	0.85	0.14	0.60
150 % NPK	0.78	0.28	0.39	0.28	0.59	0.28	0.81	0.14	0.61
100 % NPK+ (HW)	0.75	0.28	0.62	0.28	0.58	0.28	0.86	0.14	0.67
100 % NPK+ Zn	0.73	0.28	0.41	0.28	0.41	0.28	0.84	0.14	0.55
100 % NP	0.59	0.28	0.38	0.28	0.46	0.28	0.83	0.14	0.52
100 % N	0.40	0.28	0.16	0.28	0.20	0.28	0.71	0.14	0.31
100 % NPK+ FYM	1.00	0.28	1.00	0.28	1.00	0.28	0.92	0.14	0.97
100 % NPK (—S)	0.72	0.28	0.39	0.28	0.55	0.28	0.83	0.14	0.58
100 % NPK+ Lime	0.80	0.28	0.88	0.28	0.59	0.28	1.00	0.14	0.77
Control	0.47	0.28	0.27	0.28	0.36	0.28	0.88	0.14	0.43

residual effect. Plant nutrient uptake is increased, microbial and enzymatic activities in the soil are augmented which have a direct bearing on the physical, chemical and biological health of the soil (Singh et al., 2015). Long-term integrated nutrient management has been reported to improve the SYI of soybean-wheat cropping system (Choudhary et al., 2018), rice-rice (Majhi et al., 2019) and sorghum-wheat (Jadhao et al., 2019). In the present study, higher SYI in 100 % NPK + FYM indicated that the conjoint use of balanced chemical fertilizers with FYM should be preferred over the sole use of fertilizers for sustaining higher yields. Recommended NPK fertilizers supply only macronutrients to the soil, which might create nutrient imbalances in the soil over the years, thereby declining crop yields also. A higher SYI and AGY recorded with the lime application could be attributed to the ameliorating effect of lime on soil acidity by increasing pH value to 6.19 from its initial value of 5.8, thereby, increasing the microbial activity, availability of nutrients like Ca, Mg, and also the improved physical attributes of the soil (Vishwanath et al., 2020). The findings of Saha et al. (2010); Olego et al. (2021) and Ejigu et al. (2023) corroborate our results. Although 100 % NPK + HW treatment recorded considerably higher wheat grain yield than chemically weeded 100 % NPK, however, it was quite <100 % NPK + FYM. This led us to conclude that recycling the above-ground weed biomass is not adequate to meet the high nutrient requirement for higher crop yields. Poor grain yield and sustainability in control indicated that the inherent soil nutrients alone could not sustain crop yields for a longer period. However, it is still better than the continuous urea application, which created highly acidic conditions in the soil, enhanced the exchangeable Al to toxic levels (Rajneesh et al., 2018) and resulted in zero crop productivity. Acidification of soil due to long-term nitrogen fertilizer application has been reported in several studies (Liu et al., 2010; Kumari et al., 2013; Brar et al., 2015).

4.2. Soil quality indicators

Application of balanced NPK fertilizers declined BD and increased SHC and MWD of the soil significantly over control and the extent was more when FYM or lime was also applied along with chemical fertilizers. This could be attributed to the addition of organic matter that resulted in an increase in pore space and good soil aggregation (Rakshit et al., 2018). The lime application might have increased the flocculation and cementing of soil aggregates, thus improving the soil structure (Thakur et al., 2023). The highest BD was recorded in 100 % N which might be ascribed to low pH conditions, undergoing ferrolysis that might have resulted in increased clay proportion by breaking down sand particles (Chaudhary et al., 2012). The BD increased and MWD and SHC declined in the sub-surface layer as compared to the surface layer. This could be due to compaction, low organic carbon content and low root biomass in the sub-surface layer (Thakur et al., 2023). The neutralizing effect of lime on soil acidity has brought the soil pH towards neutrality (pH 6.19) (Chimdi et al., 2012). The application of FYM had a moderating effect on soil reaction, which could be ascribed to the decrease in the activity of Al^{3+} ions in the soil solution due to chelation by organic molecules (Hue, 1992). The continuous application of 100 % N alone resulted in the drastic decline in soil pH which could be due to the release of H^{+} ions during the nitrification process and enhanced exchangeable H^{+} and Al^{3+} in the soil (Condron et al., 1993; Liu et al., 2014). The addition of salts through fertilizers and solubilization of native minerals due to the reduction in pH might have resulted in a marginal increase in EC of the soil over control (Hemalatha et al., 2013). On the other hand, 100 % NPK + FYM treatment recorded significantly lower EC than fertilizer-alone treatments.

Organic and labile C improved with the balanced NPK fertilizer application over unfertilized plots due to C addition through the root

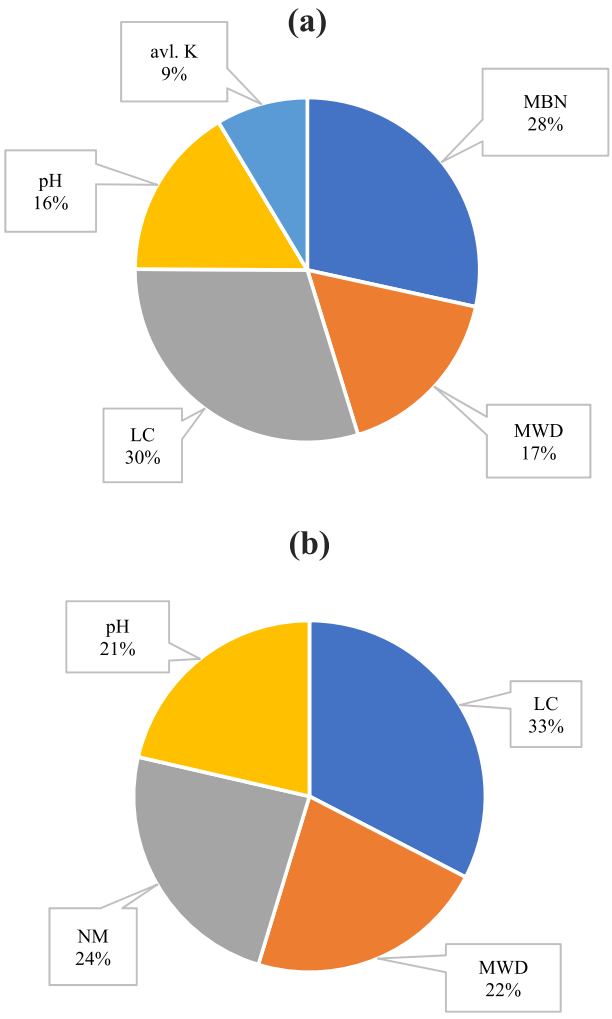


Fig. 4. Average contribution of each indicator to the soil quality index at (a) 0–0.15 m and (b) 0.15–0.30 m soil depths.

exudates, crop residues and rhizodeposition (Kumari et al., 2013). In this study, the slow rate of organic matter decomposition in the wet temperate zone could be another reason for OC build-up (Sharma et al., 2002; Verma et al., 2012). The SOC in 100 % NPK + FYM was significantly higher than all other treatments which was attributed to the direct addition of C through FYM and crop residues and root biomass (Hemalatha and Chellamuthu, 2013; Brar et al., 2015). A higher OC and LC content in 100 % NPK + HW treatment in comparison to 100 % NPK treatment might be due to the recycling of C through weed biomass in these plots. Available N, P and K content in 100 % NPK + FYM increased significantly over 100 % NPK because of the continuous addition of organic matter and nutrients in a balanced form, which might have also improved the nutrient dynamics (Verde et al., 2013; Sharma, 2015). Continuous mining of soil nutrients through intensive cropping for 47 years without external addition of NPK, PK and K led to a significant decline in their availability in control, 100%N and 100 % NPK plots, respectively, as reported earlier by Shahid et al. (2013) and Jadhao et al. (2019).

Higher DTPA-extractable Fe, Mn, Zn and Cu in FYM-amended plots than 100 % NPK plots might be due to the direct addition and solubilization of native micronutrients by organic acids produced from the decomposition of added organic sources, production of chelating agents which have the ability to reduce the adsorption of micronutrients, fixation and precipitation resulting in their enhanced availability in the soil (Dhiman et al., 2019). The Zn availability in 100 % NPK + Zn treatment was the highest among all the treatments due to the addition of Zn from $ZnSO_4$ for almost 38 years (Shambhavi et al., 2020). Significantly higher MBC, MBN and NM recorded in 100 % NPK treatment than control can be ascribed to the supply of nutrients in balanced proportion. On the other hand, FYM and lime amended plots recorded a marked improvement in the biological properties over 100 % NPK treatment because of the higher root biomass and crop residues in addition to the readily mineralizable C and N, and better growth conditions available to the soil microflora (Liu et al., 2010; Shahid et al., 2013). Incorporation of weed biomass had a positive impact on the microbial biomass as indicated in 100 % NPK + HW treatment where MBC, MBN and NM were significantly higher than in 100 % NPK treatment. Chemical fertilizers only supply mineral nutrients to the microbes, but not C which is needed for microbial cell synthesis (Singh et al., 2015). The addition of organic matter in the form of weed biomass served as the readily available C substrate thus improving the biological health of the soil. Continuous urea application in 100 % N created highly acidic conditions which might have harmed the beneficial micro-organisms, therefore declined the overall soil biological health (Meena and Sharma, 2016).

4.3. Selection of MDS

In this study, PCA was employed to select MDS for soil quality indexing. Barring EC, 16 physical, chemical and microbiological soil attributes were chosen for PCA. Since the soil was acidic and had low values of EC, it was dropped. In PC1 for 0–0.15 m soil depth, MBN, MWD and LC were selected. The MBN was chosen because of its high factor loading and correlation sum values, as suggested by Singh et al. (2015). Various researchers have ascribed the soil microbial biomass as a sensitive indicator for determining the enhancement or degradation of soil quality under different cropping practices (Liu et al., 2010; Chinnadurai et al., 2014). It directly relates to the nutrient transformation processes in the soil such as mineralization and immobilization (Dhull et al., 2004). Among physical attributes, MWD was chosen instead of SHC even though the latter had higher factor loading and correlation sum value because considering the high rainfall and risk of soil erosion in the study region, MWD can be considered a more significant parameter. The variables should be chosen by taking climate, rainfall and associated pedogenic processes of the study area into consideration (Vasu et al., 2016). The LC was selected among chemical properties instead of OC despite having a lower factor loading value (0.941) because LC is more sensitive to management practices (Kahsay et al., 2023). For PC2, only pH was selected because of its highest factor loading value (−0.825) and none of the other parameters were within the 10 % range. Considering the acidic nature of the soil, pH is a key parameter having a direct impact on the nutrient availability in the soil and thereby crop yield (Larson and Pierce, 1994). In PC3, DTPA Zn had the highest factor loading value (0.866), followed by DTPA Fe (0.280) but both of these were not chosen because they were well above the critical limit. Instead avl. K was chosen as it is an important nutrient for the growth and development of crops (Kurbah and Dixit, 2019). According to Dalal and Moloney (2000) and Andrews et al. (2002a), choice among well-correlated variables for MDS can also be based on the practicability of the variables. Hence, one can use expert opinion to retain or drop the variables from the final MDS.

For 0.15–0.30 m soil depth, LC, MWD and NM were selected out of strongly correlated nine variables for MDS in PC1. The highest factor loading value of LC led to its selection, as proposed by (Andrews et al., 2002b). The LC fraction of soil organic matter is most readily decomposable by soil microorganisms. The amount of LC in the soil has an important effect on soil biological activities such as the mineralization of nutrients and specific enzyme activities. The MWD and NM were included in the MDS despite their low factor loading value among respective physical and biological variables to bring variation in the data because of their lowest correlation sum. In PC2, the variable with the highest factor loading value i.e., pH was selected as it is a highly important indicator, which influences most of the soil functions. The

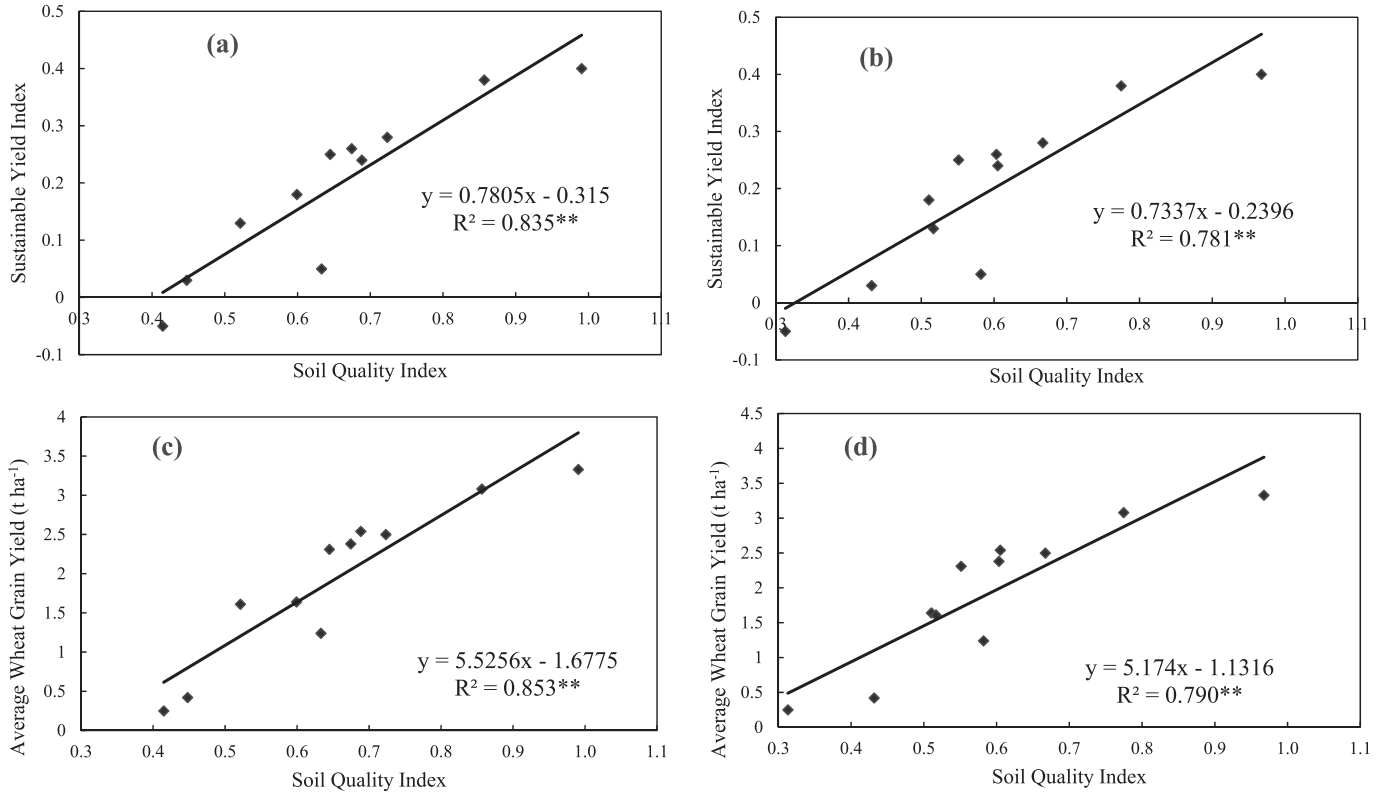


Fig. 5. The linear relationship of soil quality indices at 0–0.15 and 0.15–0.30 m soil depths with sustainable yield indices (a and b) and average wheat grain yield (c and d).

results of regression analysis between soil properties and the objective i.e., SYI also justified the inclusion of these parameters (Supplementary Table 3).

4.4. Soil quality index

Maximum SQI was recorded in 100 % NPK + FYM, followed by 100 % NPK + lime. The inclusion of FYM with the balanced NPK fertilizers enhanced the organic matter content, microbial activity and led to the accrual of residual nutrients in the soil which might have led to the higher SQI (Ram et al., 2016). The positive impact of cumulative use of organic manure and chemical fertilizers on soil quality is consistent with previously reported results (Li et al., 2013b; Singh et al., 2015). The improved SQI due to lime application might be because of the lowered Al^{3+} ions from the exchange complex (Olego et al., 2021), improved physical structure of the soil (Hynes and Naidu, 1998), increased P, Ca, Mg availability (Thakur et al., 2023), suggesting that application of either lime or FYM are quite effective to improve the quality of a continuously-cultivated acid soil. Imbalanced fertilizer application in 100 % N, 100 % NP and NPK(—S) resulted in a considerable decline in the SQI as compared to the balanced 100 % NPK treatment. Ram et al. (2016) ascertained a respective decline of 11.2, 2.7 and 0.8 % in the SQI due to the omission of PK, S and K from the fertilizer schedule. Chaudhury et al. (2005) also reported a negative impact of continuous cropping with imbalanced fertilizer use on the SQI. The SQI of plots under 50 % NPK and 100 % NPK + FYM declined with soil depth probably because of the considerable reduction in organic matter content at 0.15–0.30 m soil depth. In other treatments, an increase in the soil pH with soil depth might have led to an increase in the soil quality index.

We found that LC was the key indicator contributing most towards the soil quality index, followed by MBN, MWD, pH and avl. K content. Labile C pool is highly sensitive to soil management practices and has a prominent effect on soil functions so it is regarded as a valuable indicator of soil quality (Haynes, 2005). Labile C content might have played a positive role in the enhancement of wheat yield as it is directly related to nutrient availability and microbial activity in the soil (Mazumdar et al., 2014). Sharma et al. (2016) also found LC to be a predominant indicator of soil quality of an Alfisol under long-term fertilization. Microbial biomass N constitutes 1–6 % of the total soil N which is a dynamic source and sink of plant available N (Haynes, 2005). According to Kurbah and Dixit (2020), higher microbial biomass N indicates a higher mineralization rate of N, thereby suggesting a higher N availability to plants. Therefore, MBN is also regarded as a fine indicator of soil quality. The mean weight diameter of the water-stable aggregates contributed almost 12 % to the soil quality index and this could be supported by the fact that high rainfall (>2500 mm per annum) and topography of the study area makes it a crucial soil indicator. Chaudhury et al. (2005) also found MWD to be an important contributor to the SQI. Soil pH is an important indicator affecting the bioavailability of plant nutrients (Goulding, 2016) and most of the physicochemical processes in the soil (Neina, 2019; Amoakwah et al., 2023). Accordingly, it was one of the key indicators contributing towards the soil quality index. Available K is an essential plant nutrient that plays a key role in numerous crop physiological functions, and it contributed nearly 6 % to the SQI. A significantly high correlation of SQI with average wheat grain yield and SYI further validated the calculated SQI.

5. Conclusions

The present research aimed to compare the effects of continuous fertilizer and amendment application on wheat yield sustainability and quality of an acid Alfisol. The study revealed that physical properties, nutrient availability, microbial biomass, and activities in the soil improved with balanced fertilizer application in combination with FYM or lime. Furthermore, higher wheat grain yields were sustained for over 47 years when soil quality improved. Conversely, imbalanced fertilizer

application created nutrient imbalances within the soil, which depleted wheat yield. The calculated SQI was positively correlated with grain yield and SYI. Labile C, microbial biomass N, mean weight diameter of water-stable aggregates, soil pH, and available K emerged as the primary contributors to the improvement of soil quality. In conclusion, the conjunction of FYM or lime essential with balanced chemical fertilizers is essential for long-term productivity and improving/maintaining the quality of acidic soils of the North-western Himalayas. Sole urea application should be avoided as it can render the soil barren.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apsoil.2023.105230.

References

Amoakwah, E., Shim, J., Kim, S., Lee, Y., Kwon, S., Sangho, J., Park, S., 2023. Impact of silicate and lime application on soil fertility and temporal changes in soil properties and carbon stocks in a temperate ecosystem. Geoderma 433, 116431. https://doi.org/10.1016/j.geoderma.2023.116431.

Andrews, S.S., Karlen, D.L., Mitchell, J.P., 2002a. A comparison of soil quality indexing methods for vegetable production systems in Northern California. Agric. Ecosyst. Environ. 90 (1), 25–45. https://doi.org/10.1016/S0167-8809(01)00174-8.

Andrews, S.S., Mitchell, J.P., Mancinelli, R., Karlen, D.L., Hartz, T.K., Horwath, W.R., Pettygrove, G.S., Scow, K.M., Munk, D.S., 2002b. On-farm assessment of soil quality in California's central valley. Agron. J. 94, 12–23. https://doi.org/10.2134/agronj2002.1200.

Andrews, S.S., Karlen, D.L., Cambardella, C.A., 2004. The soil management assessment framework. Soil Sci. Soc. Am. J. 68, 1945–1962. https://doi.org/10.2136/sssaj2004.1945.

Brar, B.S., Singh, J., Singh, G., Kaur, G., 2015. Effects of long-term application of inorganic and organic fertilizers on soil organic carbon and physical properties in maize-wheat rotation. Agron 5, 220–238. https://doi.org/10.3390/agronomy5020220.

Bray, R.H., Kurtz, L.T., 1945. Determination of total, organic, and available forms of phosphorus in soils. Soil Sci. 59, 39–45. https://doi.org/10.1097/00010694-194501000-00006.

Bucher, E., 2002. Soil Quality Characterization and Remediation in Relation to Soil Management. PhD Thesis,. Department of Crop and Soil Sciences, Pennsylvania State University, USA.

Chandel, S., Hadda, M.S., Mahal, A.K., 2018. Soil quality assessment through minimum data set under different land uses of submontane Punjab. Commun. Soil Sci. Plant Anal. 49, 658–674. https://doi.org/10.1080/00103624.2018.1425424.

Chaudhary, R.S., Sinha, N.K., Mohanty, M., 2012. Effect of organics on soil aggregate stability. In: Use of Simulation Modelling in Climate Change Research: Special References to Natural Resource Management, Training Manual by ICAR Sponsored Short Course Training from 3-12th Oct, 2012. Division of Soil Physics, Indian Institute of Soil Science, Nabibagh, Berasia Road, Bhopal, India, pp. 106–118.

Chaudhury, J., Mandal, U.K., Sharma, K.L., Ghosh, H., Mandal, B., 2005. Assessing soil quality under long-term rice-based cropping system. Commun. Soil Sci. Plant Anal. 36 (9–10), 1141–1161. https://doi.org/10.1081/CSS-200056885.

Chimdi, A., Gebrekidan, V., Kibret, K., Tadesse, A., 2012. Effects of liming on acidity-related chemical properties of soils of different land use systems in Western Oromia, Ethiopia. World J. Agric. Sci. 8 (6), 560–567. https://doi.org/10.5829/idosi.wjas.2012.8.6.1686.

Chinnadurai, C., Gopalaswamy, G., Balachandar, D., 2014. Impact of long-term organic and inorganic nutrient managements on the biological properties and eubacterial community diversity of the Indian semi-arid Alfisol. Arch. Agron. Soil Sci. 60 (4), 531–548. https://doi.org/10.1080/03650340.2013.803072.

Choudhary, M., Panday, S.C., Meena, V.S., Singh, V., Yadav, R.P., Mahanta, D., Mondal, T., Mishra, P.K., Bisht, J.K., Pattanayak, A., 2018. Long-term effects of organic manure and inorganic fertilization on sustainability and chemical soil quality indicators of soybean-wheat cropping system in the Indian mid-Himalayas. Agric. Ecosyst. Environ. 257, 38–46. https://doi.org/10.1016/j.agee.2018.01.029.

Condron, L.M., Tiessen, H., Trasar-Cepeda, C., Moir, J.O., Stewart, J.W.B., 1993. Effects of liming on organic matter decomposition and phosphorus extractability in an acid humic Ranker soil from northwest Spain. Biol. Fertil. Soils 15, 279–284. https://doi.org/10.1007/BF00337213.

Dalal, R.C., Moloney, D., 2000. Sustainability indicators of soil health and biodiversity. In: Hale, P., Petrie, A., Moloney, D., Sattler, P. (Eds.), Management for Sustainable Ecosystems. Centre for Conservation Biology, The University of Queensland, Brisbane, pp. 101–108.

Dhiman, D., Sharma, R.P., Sankhyan, N.K., Sepehya, S., Sharma, S.K., Kumar, R., 2019. Effect of regular application of fertilizers, manure and lime on soil health and productivity of wheat in an acid Alfisol. J Pl. Nutr. 42 (19), 2507–2521. https://doi.org/10.1080/01904167.2019.1659317.

Dhull, S., Goyal, S., Kapoor, K., Mundra, M., 2004. Microbial biomass carbon and microbial activities of soils receiving chemical fertilizers and organic amendments. Arch. Agron. Soil Sci. 50 (6), 641–647. https://doi.org/10.1080/08927010400011294.

Dutta, J., Sharma, S.P., Sharma, S.K., Sharma, G.D., Sankhyan, N.K., 2015. Indexing soil quality under long-term maize-wheat cropping system in an acidic Alfisol. Commun. Soil Sci. Plant Anal. 46 (15), 1841–1862. https://doi.org/10.1080/00103624.2015.1047845.

Ejigu, W., Selassie, Y.G., Elias, E., Molla, E., 2023. Effect of lime rates and method of application on soil properties of acidic Luvisols and wheat (*Triticum aestivum* L.) yields in northwest Ethiopia. Heliyon 9 (3), e13988. https://doi.org/10.1016/j.heliyon.2023.e13988.

Erkossa, T., Itanna, F., Stahr, K., 2007. Indexing soil quality: a new paradigm in soil science research. Aust. J. Soil Res. 45, 129–137. https://doi.org/10.1071/SR06064.

Goulding, K.W.T., 2016. Soil acidification and the importance of liming agricultural soils with particular reference to the United Kingdom. Soil Use Manag. 32 (3), 390–399. https://doi.org/10.1111/sum.12270.

Gourav, N., Sankhyan, N.K., Sharma, R.P., Sharma, G.D., 2018. Vertical distribution of sulfur fractions in a continuously fertilized acid Alfisol under maize–wheat cropping system. Commun. Soil Sci. Plant Anal. 49 (8), 923–933. https://doi.org/10.1080/00103624.2018.1448409.

Haynes, R.J., 2005. Labile organic matter fractions as central components of the quality of agricultural soils: a review. Adv. Agron. 5, 221–268. https://doi.org/10.1016/S0065-2113(04)85005-3.

Hemalatha, S., Chellamuthu, S., 2013. Impact of long-term fertilization on soil nutritional quality under finger millet-maize cropping sequence. J. Environ. Res. Dev. 7, 1571–1576.

Hemalatha, S., Radhika, K., Maragatham, S., Katharine, S.P., 2013. Influence of long-term fertilization on soil fertility. J. Agric. Allied Sci. 2, 30–36.

Hue, N.V., 1992. Correcting soil acidity of a highly weathered Ultisols with chicken manure and sewage sludge. Commun. Soil Sci. Plant Anal. 23, 241–264. https://doi.org/10.1080/00103629209368586.

Hynes, R.J., Naidu, R., 1998. Influence of lime, fertilizer and manure applications on soil organic matter content and soil physical conditions: a review. Nutr. Cycl. Agroecosyst. 51, 123–137. https://doi.org/10.1023/A:1009738307837.

Jackson, M.L., 1973. Soil Chemical Analysis. Prentice Hall of India Pvt. Ltd.

Jadhao, S.D., Mali, D.V., Kharche, V.K., Singh, M., Bhoyar, S.M., Kadu, P.R., Wanjari, R. H., Sonune, B.A., 2019. Impact of continuous manuring and fertilization on changes in soil quality under sorghum-wheat sequence on a vertisols. J. Indian Soc. Soil Sci. 67 (1), 55–64. https://doi.org/10.5958/0974-0228.2019.00006.9.

Jenkinson, D.S., Ladd, J.N., 1981. Microbial biomass in soil: measurement and turnover. In: Paul, E.A., Ladd, J.N. (Eds.), Soil Biochemistry. Marcel Dekker Inc., New York, pp. 415–457.

Ji, L., Ni, K., Wu, Z., Zhang, J., Yi, X., Yang, X., Ling, N., You, Z., Guo, S., Ruan, J., 2020. Effect of organic substitution rates on soil quality and fungal community composition in a tea plantation with long-term fertilization. Biol. Fertil. Soils 56, 633–646. https://doi.org/10.1007/s00374-020-01439-y.

Kahsay, A., Haile, M., Gebresamuel, G., Mohammed, M., Okolo, C.C., 2023. Assessing land use type impacts on soil quality: application of multivariate statistical and expert opinion-followed indicator screening approaches. Catena 231, 107351. https://doi.org/10.1016/j.catena.2023.107351.

Kaiser, H.F., 1960. The application of electronic computers to factor analysis. Educ. Psychol. Meas. 20 (1), 141–151. https://doi.org/10.1177/001316446002000116.

Keeney, D.R., 1982. Nitrogen-availability indices. In: Page, A.L., Miller, R.H., Keeney, D. R. (Eds.), Methods in Soil Analysis - Part 2 Chemical and Microbial Properties, second ed. Soil Science Society of America Inc., Publisher Madison, Wisconsin, USA, pp. 711–733.

Klute, A., 1965. Laboratory measurement of hydraulic conductivity of saturated soil. In: Black, C.A. (Ed.), Methods of Soil Analysis. Part 1. American Society of Agronomy, Inc., Madison, WI, pp. 210–211. https://doi.org/10.2134/agronmonogr9.1.c13.

Kumari, G., Thakur, S.K., Kumar, N., Mishra, B., 2013. Long-term effect of fertilizer, manure and lime on yield sustainability and soil organic carbon status under maize (*Zea mays*) - wheat (*Triticum aestivum*) cropping system in Alfisols. Indian J. Agron. 58, 152–158.

Kurbah, I., Dixit, S.P., 2019. Soil potassium fractions as influenced by integrated fertilizer application based on soil test crop response under maize-wheat cropping systems in acid Alfisol. Int. J. Econ. Pl. 6 (1), 25–29. https://doi.org/10.23910/ijep/2019.6.1.0291.

Kurbah, I., Dixit, S.P., 2020. Effect of fertilizer application based on soil test crop response approach on active and passive pools of soil organic matter in maize–wheat cropping system. Arch. Agron. Soil Sci. 66 (6), 858–871. https://doi.org/10.1080/03650340.2019.1642468.

Larson, W.E., Pierce, F.J., 1994. The dynamics of soil quality as a measure of sustainable management. In: Doran, J.W., Coleman, D.C., Bezdicek, D.F., Stewart, B.A. (Eds.), Defining Soil Quality for a Sustainable Environment. https://doi.org/10.2136/sssaspecpub35.c3.

Li, P., Zhang, T., Wang, X., Yu, D., 2013a. Development of biological soil quality indicator system for sub-tropical China. Soil Tillage Res. 126, 112–118. https://doi.org/10.1016/j.still.2012.07.011.

Li, Q., Xu, M., Liu, G., Zhao, Y., Tuo, D., 2013b. Cumulative effects of a 17-year chemical fertilization on the soil quality of cropping system in the Loess Hilly Region, China. J. Plant Nutr. Soil Sci. 176 (2), 249–259. https://doi.org/10.1002/jpln.v176.2.

Lindsay, W.L., Norvell, W.A., 1978. Development of a DTPA soil test for zinc, iron, manganese and copper. Soil Sci. Soc. Am. J. 42, 421–428. https://doi.org/10.2136/sssaj1978.03615995004200030009x.

Liu, E., Yan, C., Mei, X., He, W., Bing, S.H., Ding, L., Liu, Q., Liu, S., Fan, T., 2010. Long term effect of chemical fertilizer, straw and manure on soil chemical and biological properties in North West China. Geoderma 158, 173–180. https://doi.org/10.1016/j.geoderma.2010.04.029.

Liu, X., Zhou, J., Li, W., Xu, J., Brookes, P.C., 2014. The combined effects of urea application and simulated acid rain on soil acidification and microbial community structure. Environ. Sci. Pollut. Res. 21, 6623–6631. https://doi.org/10.1007/s11356-014-2573-9.

Majhi, P., Rout, K.K., Nanda, G., Singh, M., 2019. Soil quality for rice productivity and yield sustainability under long-term fertilizer and manure application. Commun. Soil Sci. Plant Anal. 50 (11), 1330–1343. https://doi.org/10.1080/00103624.2019.1614607.

Mazumdar, S.P., Kundu, D.K., Ghosh, D., Saha, A.R., Majumdar, B., Ghorai, A.K., 2014. Effect of long-term application of inorganic fertilizers and organic manure on yield, potassium uptake and distribution of potassium fractions in the New Gangetic alluvial soil under jute-rice-wheat cropping system. Int. J. Agric. Sci. Food Technol. 5, 297–306.

Mbuthia, L.W., Acosta-Martínez, V., DeBryun, J., Schaeffer, S., Tyler, D., Odoi, E., Mpheshea, M., Walker, F., Eash, N., 2015. Long-term tillage, cover crop, and fertilization effects on microbial community structure, activity: implications for soil quality. Soil Biol. Biochem. 89, 24–34. https://doi.org/10.1016/j.soilbio.2015.06.016.

Meena, H.M., Sharma, R.P., 2016. Long-term effect of fertilizers and amendments on different fractions of organic matter in an acid Alfisol. Commun. Soil Sci. Plant Anal. 47 (11), 1430–1440. https://doi.org/10.1080/00103624.2016.1178766.

Mukherjee, A., Lal, R., 2014. Comparison of soil quality index using three methods. PLoS One 9 (8), e105981. https://doi.org/10.1371/journal.pone.0105981.

Neina, D., 2019. The role of soil pH in plant nutrition and soil remediation. Appl. Environ. Soil Sci. 5794869 https://doi.org/10.1155/2019/5794869.

Ning, C.C., Gao, P.D., Wang, B.Q., Lin, W.P., Jiang, N.H., Cai, K.Z., 2017. Impacts of chemical fertilizer reduction and organic amendments supplementation on soil nutrient, enzyme activity and heavy metal content. J. Integr. Agric. 16 (8), 1819–1831. https://doi.org/10.1016/S2095-3119(16)61476-4.

Olego, M.Á., Quiroga, M.J., Mendaña-Cuervo, C., Cara-Jiménez, J., López, R., Garzón-Jimeno, E., 2021. Long-term effects of calcium-based liming materials on soil fertility sustainability and rye production as soil quality indicators on a Typic Palexerult. Processes 9 (7), 1181. https://doi.org/10.3390/pr9071181.

Prasad, R., 2005. Rice-wheat cropping system. Adv. Agron. 86, 255–339. https://doi.org/10.1016/S0065-2113(05)86006-7.

Rajneesh, S., Sharma, R.P., Sankhyan, N.K., Kumar, R., 2017. Long-term effect of fertilizers and amendments on depth-wise distribution of available NPK, micronutrient cations, productivity, and NPK uptake by maize-wheat system in an acid Alfisol of northwestern Himalayas. Commun. Soil Sci. Plant Anal. 48 (18), 2193–2209. https://doi.org/10.1080/00103624.2017.1408816.

Rajneesh, Sharma, R.P., Sankhyan, N.K., Kumar, R., Sepehya, S., 2018. Effect of a four-decade long application of fertilizers, farmyard manure and lime, on forms of soil acidity and their relationship with yield of wheat and maize in an acid Alfisol. J. Plant Nutr. 41 (11), 1444–1455. https://doi.org/10.1080/01904167.2018.1457684.

Rakshit, R., Das, A., Padbhushan, R., Sharma, R.P., Sushant, Kumar, S., 2018. Assessment of soil quality and identification of parameters influencing system yield under long-term fertilizer trial. J. Indian Soc. Soil Sci. 66, 166–171.

Ram, S., Singh, V., Sirari, P., 2016. Effects of 41 years of application of inorganic fertilizers and farm yard manure on crop yields, soil quality, and sustainable yield index under a rice-wheat cropping system on Mollisols of north India. Commun. Soil Sci. Plant Anal. 47 (2), 179–193. https://doi.org/10.1080/00103624.2015.1109653.

Saha, R., Mishra, V.K., Majumdar, B., Laxminarayana, K., Ghosh, P.K., 2010. Effect of integrated nutrient management on soil physical properties and crop productivity under a maize (*Zea mays*)–mustard (*Brassica campestris*) cropping sequence in acidic soils of northeast India. Commun. Soil Sci. Plant Anal. 41 (18), 2187–2200. https://doi.org/10.1080/00103624.2010.504799.

Saha, S., Saha, B.N., Ray, M., Mukhopadhyay, S.K., Halder, P., Das, A., Chatterjee, S., Pramanick, M., 2018. Integrated nutrient management (INM) on yield trends and sustainability, nutrient balance and soil fertility in a long-term (30 years) rice-wheat system in the Indo-Gangetic plains of India. J. Plant Nutr. 41, 2365–2375. https://doi.org/10.1080/01904167.2018.1510509.

Saha, S., Saha, B., Mohanty, S., Ray, M., Hazra, G.C., Chatterjee, S., Mukhopadhyay, S.K., 2022. Identifying suitable soil quality indicators in an Indian rice-wheat system as influenced by long-term (30 years) inorganic-organic fertilization. Arch. Agron. Soil Sci. 68 (5), 704–718. https://doi.org/10.1080/03650340.2020.1851684.

Shahid, M., Nayak, A.K., Shukla, A.K., Tripathi, R., Kumar, A., Mohanty, S., Bhattacharyya, P., Raja, R., Panda, B.B., 2013. Long-term effects of fertilizer and manure applications on soil quality and yields in a sub-humid tropical rice-rice system. Soil Use Manag. 29, 322–332. https://doi.org/10.1111/sum.12050.

Shahid, M., Shukla, A.K., Bhattacharyya, P., Tripathi, R., Mohanty, S., Kumar, A., Lal, B., Gautam, P., Raja, R., Panda, B.B., Das, B., Nayak, A.K., 2016. Micronutrients (Fe, Mn, Zn and Cu) balance under long-term application of fertilizer and manure in a tropical rice–rice system. J. Soils Sediments 16, 737–747. https://doi.org/10.1007/s11368-015-1272-6.

Shambhavi, S., Kumar, R., Padbhushan, R., Verma, G., Sharma, S.P., Sharma, R.P., 2020. Dynamics of zinc under long-term application of chemical fertilizers and amendments by maize–wheat cropping sequence in Typic Hapludalfs. Soil Use Manag. 36, 507–523. https://doi.org/10.1111/sum.12566.

Sharma, T., 2015. Response of integrated nutrient supply on yield of wheat and physical-chemical properties of soil. Bioscan 10, 77–80.

Sharma, S.P., Subehia, S.K., Sharma, P.K., 2002. Long-Term Effects of Chemical Fertilizers on Soil Quality, Crop Productivity and Sustainability. Research Bulletin, CSK Himachal Pradesh Krishi Vishvavidyalaya, Palampur, India, p. 33.

Sharma, K.L., Ramachandrappa, B.K., Chandrika, D.S., Sathish, A., Dhanpal, G.N., Rao, Ch.S., Shankar, M.A., Grace, J.K., Sankar, G.R.M., Chary, G.R., Munnalal, T., Kumar, S., Rani, K.U., Venkateswarlu, B., 2016. Effect of organic manure and crop residue-based long-term nutrient management systems on soil quality changes under sole finger millet (Eleusine coracana (L.) Gaertn.) and groundnut (Arachis hypogaea L.) - finger millet rotation in rainfed Alfisol Commun. Soil Sci. Plant Anal. 47 (7), 899–914. https://doi.org/10.1080/00103624.2016.1147045.

Sheoran, O.P., Tonk, D.S., Kaushik, L.S., Hasija, R.C., Pannu, R.S., 1998. Statistical software package for agricultural research workers. In: Hooda, D.S., Hasija, R.C. (Eds.), Recent Advances in Information Theory, Statistics & Computer Applications. Department of Mathematics Statistics, CCS HAU, Hisar, India. http://14.139.232.166/opstat/.

Shukla, M.K., Lal, R., Ebinger, M., 2006. Determining soil quality indicators by factor analysis. Soil Tillage Res. 87, 194–204. https://doi.org/10.1016/j.still.2005.03.011.

Singh, R.A., 1980. Soil Physical Analysis. Kalyani Publishers, Ludhiana, India.

Singh, M.V., 2007. Problems of micro- and secondary nutrients in acid soils of India and their management. Bull. Indian Soc. Soil Sci. 25, 27–58.

Singh, R.P., Das, S.K., Rao, U.M.B., Reddy, M.N., 1990a. Towards Sustainable Dryland Agriculture Practices. Bulletin, CRIDA, Hyderabad, India, pp. 5–9.

Singh, R.P., Das, S.K., Rao, U.M.B., Reddy, M.N., 1990b. Sustainability Index under Different Management: Annual Report. CRIDA, Hyderabad, India.

Singh, S.R., Kundu, D.K., Tripathi, M.K., Dey, P., Saha, A.R., Kumar, M., Singh, I., Mahapatra, B.S., 2015. Impact of balanced fertilization on nutrient acquisition, fibre yield of jute and soil quality in New Gangetic alluvial soils of India. Appl. Soil Ecol. 92, 24–34. https://doi.org/10.1016/j.apsoil.2015.03.007.

Soil Survey Staff, 1975. Soil Taxonomy. Agriculture Handbook. USDA, p. 436.

Subbiah, B.V., Asija, G.L., 1956. A rapid procedure for the determination of available nitrogen in soils. Curr. Sci. 25, 254–260.

Thakur, P., Paliyal, S.S., Dev, P., Datt, N., 2022a. Methods and approaches - soil quality indexing, minimum data set selection & interpretation - a critical review. Commun. Soil Sci. Plant Anal. 53 (15), 1849–1864. https://doi.org/10.1080/00103624.2022.2063328.

Thakur, A., Sharma, R.P., Sankhyan, N.K., Anjali., 2022b. Long term effect of fertilizers and amendments on macronutrients uptake by maize and relationship with soil organic carbon in maize-wheat system in acid Alfisol of north-western Himalayas. Indian J. Ecol. 49 (2), 388–393. https://doi.org/10.55362/IJE/2022/3534.

Thakur, A., Sharma, R.P., Sankhyan, N.K., Sepehya, S., 2023. Effect of 46 years' application of fertilizers, FYM and lime on physical, chemical and biological properties of soil under maize–wheat system in an acid Alfisol of northwest Himalayas. Soil Use Manag. 39, 357–367. https://doi.org/10.1111/sum.12821.

Vance, E.D., Brookes, P.C., Jenkinson, D.S., 1987. Microbial biomass measurements in forest soils: the use of the chloroform fumigation incubation method for strongly acid soils. Soil Biol. Biochem. 19, 697–702. https://doi.org/10.1016/0038-0717(87)90051-4.

Vasu, D., Singh, S.K., Ray, S.K., Duraisami, P.V., Tiwary, P., Chandran, P., Nimkar, A.M., Anantwar, S.G., 2016. Soil quality index (SQI) as a tool to evaluate crop productivity in semi-arid Deccan plateau, India. Geoderma 282, 70–79. https://doi.org/10.1016/j.geoderma.2016.07.010.

Verde, B.S.D., Oginga, B., Njeri, M.J., 2013. Effects of manure, lime and mineral P fertilizer on soybean yields and soil fertility in a humic Nitisol in the Central Highlands of Kenya. Int. J. Agric. Sci. Res. 2, 283–291.

Verma, G., Sharma, R.P., Sharma, S.P., Subehia, S.K., Shambhavi, S., 2012. Changes in soil fertility status in maize-wheat system due to long-term use of chemical fertilizers and amendments in an Alfisol. Plant Soil Environ. 58, 529–533. https://doi.org/10.17221/133/2012-PSE.

Vishwanath, S.K., Purakayastha, T.J., Datta, S.P., Rosin, K.G., Mahapatra, P., Sinha, S.K., Yadav, S.P., 2020. Impact of forty-seven years of long-term fertilization and liming on soil health, yield of soybean and wheat in an acidic Alfisol. Arch. Agron. Soil Sci. 68 (4), 531–546. https://doi.org/10.1080/03650340.2020.1843023.

Walkley, A., Black, I.A., 1934. An examination of the Degtjareff (wet acid) method for determining soil organic matter and a proposed modification of the chromic acid

titration method. Soil Sci. 37 (1), 29–38. https://doi.org/10.1097/00010694-193401000-00003.
Wander, M.M., Bollero, G.A., 1999. Soil quality assessment of tillage impacts on Illinois. Soil Sci. Soc. Am. J. 63 (4), 961–971. https://doi.org/10.2136/sssaj1999.634961x.
Waring, S., Bremner, J., 1964. Ammonium production in soil under waterlogged conditions as an index of nitrogen availability. Nature 201, 951–952. https://doi.org/10.1038/201951a0.
Weil, R., Islam, K., Stine, M., Gruver, J., Samson-Liebig, S., 2003. Estimating active carbon for soil quality assessment: a simplified method for laboratory and field use. Am. J. Altern. Agric. 18 (1), 3–17. https://doi.org/10.1079/AJAA200228.
Williams, A., Börjesson, G., Hedlund, K., 2013. The effects of 55 years of different inorganic fertiliser regimes on soil properties and microbial community composition. Soil Biol. Biochem. 67, 41–46. https://doi.org/10.1016/j.soilbio.2013.08.008.
Yang, Z.C., Zhao, N., Huang, F., Lv, Y.Z., 2015. Long-term effects of different organic and inorganic fertilizer treatments on soil organic carbon sequestration and crop yields on the North China Plain. Soil Tillage Res. 146, 47–52. https://doi.org/10.1016/j.still.2014.06.011.
Yoder, R.E., 1936. A direct method of aggregate analysis and study of the physical nature of erosion losses. J. Am. Soc. Agron. 28, 337–351. https://doi.org/10.2134/agronj1936.00021962002800050001x.
Zhou, M., Xiao, Y., Li, Y., Zhang, X., Wang, G., Jin, J., Ding, G., Liu, X., 2022. Soil quality index evaluation model in responses to six-year fertilization practices in Mollisols. Arch. Agron. Soil Sci. 68 (2), 180–194. https://doi.org/10.1080/03650340.2020.1827395.